

Instructions

- Attempt all questions
- Write down your roll# on the question paper.
- Attach your question paper with the answer sheet before leaving examination hall.

Part I

Consider the following rule for Part I

$\langle \text{while} \rangle \rightarrow \text{while}(\langle \text{condition} \rangle) \{ \langle \text{block} \rangle \langle \text{change} \rangle \}$
 $\langle \text{block} \rangle \rightarrow \langle \text{while} \rangle \mid x$
 $\langle \text{dec} \rangle \rightarrow \langle \text{type} \rangle \langle \text{list} \rangle;$
 $\langle \text{type} \rangle \rightarrow \text{int} \mid \text{char} \mid \text{float} \mid \text{string}$
 $\langle \text{list} \rangle \rightarrow \text{id}(\text{E} := \text{val}) \mid \text{id}(\text{E} := \text{val}), \langle \text{list} \rangle$
 $\langle \text{condition} \rangle \rightarrow 0 \mid 1$
 $\langle \text{change} \rangle \rightarrow \text{id} \langle \text{in} \rangle$
 $\langle \text{in} \rangle \rightarrow ++ \mid --$

Question #01 CLO-2 (12)

Construct the Deterministic flow to construct shift reduced parsing

Question #02 CLO-3 (05+05)

- Construct LR (0) Parsing table for the given rule.
- Construct SLR(1) Parsing table for the given rule.

Question #03 CLO-3 (08)

Perform stack processing for any of the sentence if possible using LR (0) parsing table.

Part II

Consider the following Rule for Part II

$T \rightarrow \text{ZZZahaZaZZ} \mid \text{starWS}$
 $W \rightarrow \text{SsSW room S flag S cloud} \mid \text{deep}$
 $S \rightarrow \text{SSrSr} \mid \text{SsS} \mid \text{SrW}$

Question #04 CLO-02 (05+10)

- Remove the left recursion from the above rule if any.
- Write first() and follow() for all non-terminals.
 - Write LL(1) Predictive parsing table for the given rule.

Question #05 CLO-03 (05)

Perform Stack Processing activity for any of the possible supported sentence

Its not the only examination of your life...